

Climate 5000 Series - Gen 4

12K BTU Single Zone Max Performance System

4-Way Cassette IDU

Engineering
Submittal
Sheet



BOSCH

Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

12K Btu Single Zone Max Performance System - 4-Way Cassette IDU

Indoor part number	8-733-962-717
Indoor model number	BMS500-AAU012-1AHCXD
Outdoor part number	8-733-962-689
Outdoor model number	BMS500-AAS012-1CSXHD

Efficiency Data		AHRI 210/240 2023
SEER2		22.2
EER2		12.5
HSPF2 Region IV		11.6
HSPF2 Region V		8.9
COP(47F)		3.70
COP (5F)		2.00

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3800~15100
	Rated Capacity	12000
	Current	4.6
	Input Power	945
Heating	Capacity Range	5300~16200
	Rated Capacity (47F)	12000
	Rated Capacity (5F)	11500
	Current	4.4
	Input Power	951

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 15.0
Max. fuse indoor/outdoor (A)	15.0 / 15.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 350
Input (W)	885
Rated current (RLA/A)	6.0

Piping	
Refrigerant precharge (oz/kg)	35.27 / 1.00
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	1/4" / Ø6.35
Gas side (in/mm)	3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.16 / 15
Max. refrigerant pipe length (ft/m)	82.00 / 25
Max. difference in height (ft/m)	49.20 / 15

Operation Range

ODU	Cooling (°F / °C)	-22~122 / -30~50
	Heating (°F / °C)	-22~75 / -30~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data		
Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		3.06
IDU	Fan motor model	ZKFN-45-8-2
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan speed (RPM)	700 / 600 / 400
	Air flow, Turbo / H / M / L / Si (CFM)	417.9 / 364.9 / 306.1 / 194.2 / 147.2
	Approx. Air Throw Range ft (m)	9.8 / 3
	Noise level, Turbo / H / M / L / Si (dB(A))	46 / 43.5 / 40.5 / 27.5 / 26.00
	Net weight (lbs / kg)	35.27 / 16.0
	Gross weight (lbs / kg)	41.89 / 19.0
	Panel net weight (lbs / kg)	5.95 / 2.7
ODU	Gross panel weight (lbs / kg)	9.48 / 4.3
	Drain pipe diameter (mm)	ODØ25mm
	Fan motor model	ZKFN-34-10-1-3
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed (RPM)	780 / 680 / 580
	Air flow, Max (CFM)	1471.5
	Noise level, Max (dB(A))	57.0
	Net weight (lbs / kg)	72.75 / 33.0
	Gross weight (lbs / kg)	78.7 / 35.7

Accessories	
BMS-CCG-XXD (8733962708)	Decorative Grille - 9K ~ 18K
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

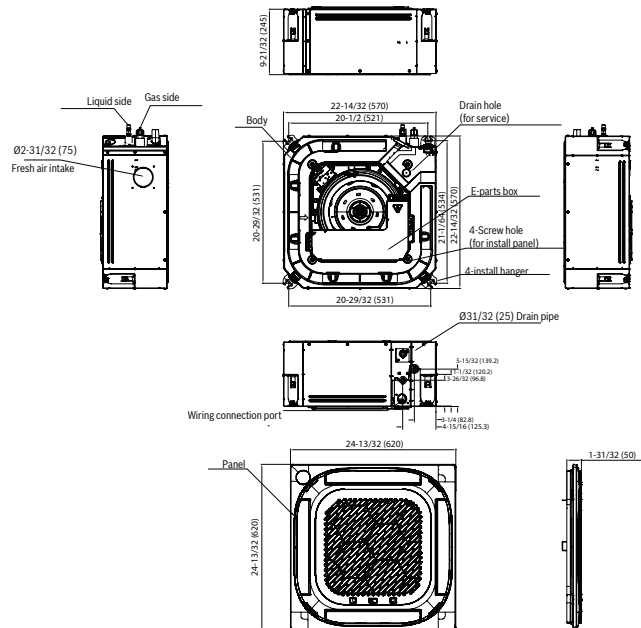
Standard Parts	
8-733-969-065	Remote Control

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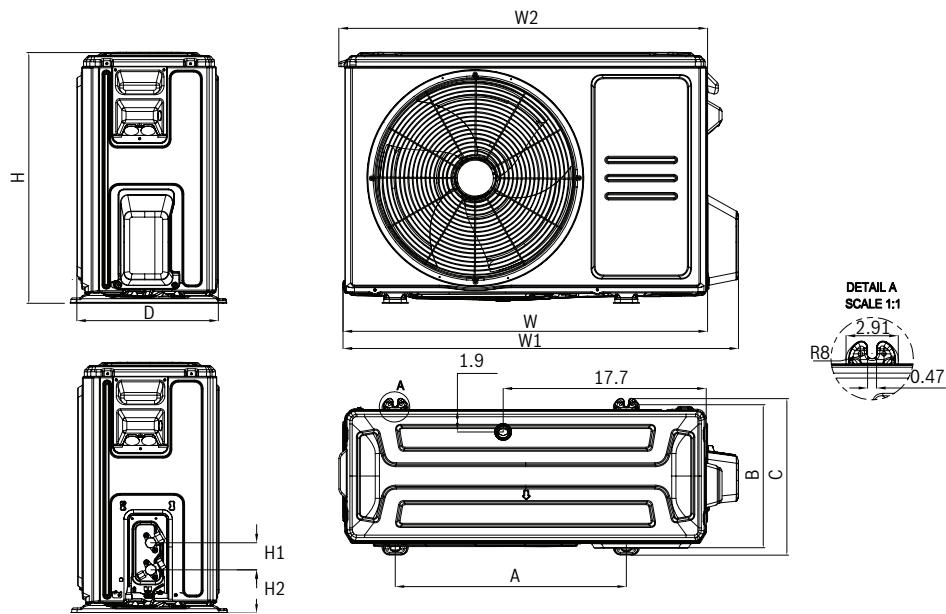


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU012-1AHCXD	22.44	22.44	9.65

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS012-1CSXHD	31.69	12.99	21.81	34.4	32.1	20.1	12.5	13.6	2.4	3.8

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Cooling Performance - Indoor Dry Bulb Temperature at 418 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	8594	8606	8615	8616	8627	9112	9123	9132	9134	9145	9657	9669	9680	9682	9693	10239	10250	10261	10262	10275
	S/T	0.60	0.71	0.80	0.80	0.91	0.57	0.67	0.76	0.76	0.86	0.53	0.63	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.77
	PI	0.37	0.38	0.38	0.38	0.38	0.40	0.40	0.40	0.40	0.40	0.42	0.42	0.42	0.42	0.42	0.45	0.45	0.45	0.45	0.45
-13°F (-25.0°C)	TC	9311	9323	9333	9334	9346	9872	9883	9893	9895	9907	10461	10475	10487	10489	10500	11092	11104	11116	11118	11131
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.39	0.39	0.39	0.39	0.39	0.41	0.41	0.41	0.41	0.41	0.44	0.44	0.44	0.44	0.44	0.46	0.46	0.46	0.46	0.46
-4°F (-20.0°C)	TC	10027	10040	10051	10051	10065	10631	10644	10654	10656	10669	11266	11281	11293	11295	11308	11945	11958	11971	11973	11988
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.79
	PI	0.40	0.40	0.40	0.40	0.40	0.43	0.43	0.43	0.43	0.43	0.45	0.45	0.45	0.45	0.45	0.48	0.48	0.48	0.48	0.48
0°F (-17.8°C)	TC	10345	10359	10370	10371	10385	10968	10981	10992	10994	11008	11624	11639	11652	11654	11667	12324	12338	12351	12353	12368
	S/T	0.63	0.73	0.84	0.84	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	0.41	0.41	0.41	0.41	0.41	0.43	0.43	0.43	0.43	0.43	0.46	0.46	0.46	0.46	0.46	0.49	0.49	0.49	0.49	0.49
5°F (-15.0°C)	TC	10743	10757	10769	10769	10784	11390	11404	11415	11417	11431	12071	12087	12100	12102	12116	12798	12812	12826	12828	12844
	S/T	0.63	0.74	0.84	0.84	0.96	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81
	PI	0.42	0.42	0.42	0.42	0.42	0.44	0.44	0.44	0.44	0.44	0.47	0.47	0.47	0.47	0.47	0.50	0.50	0.50	0.50	0.50
14°F (-10.0°C)	TC	10995	11010	11022	11022	11037	11658	11672	11683	11685	11699	12354	12370	12384	12386	12400	13099	13113	13127	13129	13146
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79
	PI	0.43	0.43	0.43	0.43	0.43	0.45	0.45	0.45	0.45	0.45	0.48	0.48	0.48	0.48	0.48	0.51	0.51	0.51	0.51	0.51
17°F (-8.3°C)	TC	11079	11094	11106	11107	11122	11747	11761	11772	11775	11789	12449	12465	12479	12481	12495	13199	13213	13228	13230	13246
	S/T	0.62	0.72	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.88	0.55	0.64	0.73	0.73	0.83	0.52	0.61	0.69	0.69	0.79
	PI	0.43	0.43	0.43	0.43	0.43	0.46	0.46	0.46	0.46	0.46	0.48	0.48	0.48	0.48	0.48	0.51	0.51	0.51	0.51	0.51
23°F (-5.0°C)	TC	11109	11124	11136	11137	11152	11779	11793	11804	11807	11821	12483	12499	12513	12515	12529	13235	13249	13264	13266	13282
	S/T	0.61	0.72	0.81	0.81	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.46	0.46	0.46	0.46	0.46	0.49	0.49	0.49	0.49	0.49	0.52	0.52	0.52	0.52	0.52	0.55	0.55	0.55	0.55	0.55
35°F (1.7°C)	TC	11170	11185	11197	11197	11212	11843	11857	11869	11871	11885	12550	12567	12581	12583	12597	13307	13321	13336	13338	13354
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	0.52	0.52	0.52	0.52	0.52	0.55	0.55	0.55	0.55	0.55	0.59	0.59	0.59	0.59	0.59	0.62	0.62	0.62	0.62	0.62
41°F (5.0°C)	TC	11659	11674	11687	11687	11703	12361	12376	12388	12391	12405	13100	13117	13131	13134	13149	13889	13904	13919	13921	13939
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.69	0.69	0.78
	PI	0.55	0.55	0.55	0.55	0.55	0.58	0.58	0.58	0.58	0.58	0.61	0.61	0.61	0.61	0.61	0.65	0.65	0.65	0.65	0.65
47°F (8.3°C)	TC	12148	12164	12177	12178	12194	12880	12895	12908	12910	12926	13649	13667	13682	13685	13700	14472	14487	14503	14505	14523
	S/T	0.63	0.73	0.83	0.83	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	0.57	0.57	0.57	0.57	0.57	0.60	0.60	0.60	0.60	0.61	0.64	0.64	0.64	0.64	0.64	0.68	0.68	0.68	0.68	0.68
50°F (10.0°C)	TC	12155	12171	12184	12185	12201	12887	12902	12915	12918	12933	13657	13675	13690	13693	13708	14480	14496	14512	14514	14532
	S/T	0.62	0.73	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.89	0.55	0.65	0.73	0.73	0.84	0.52	0.61	0.69	0.69	0.79
	PI	0.57	0.57	0.57	0.57	0.57	0.61	0.61	0.61	0.61	0.61	0.64	0.64	0.64	0.64	0.64	0.68	0.68	0.68	0.68	0.68
59°F (15.0°C)	TC	12176	12192	12205	12206	12222	12910	12925	12938	12940	12956	13681	13699	13714	13717	13732	14505	14521	14537	14539	14557
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	0.58	0.58	0.58	0.58	0.58	0.62	0.62	0.62	0.62	0.62	0.65	0.65	0.65	0.66	0.66	0.69	0.69	0.69	0.69	0.70
68°F (20.0°C)	TC	12197	12214	12227	12227	12244	12932	12948	12960	12963	12978	13705	13723	13738	13740	13756	14531	14546	14562	14564	14583
	S/T	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78	0.48	0.57	0.65	0.65	0.74
	PI	0.59	0.59	0.59	0.59	0.59	0.63	0.63	0.63	0.63	0.63	0.66	0.66	0.67	0.67	0.67	0.70	0.70	0.71	0.71	0.71
77°F (25.0°C)	TC	12218	12235	12248	12249	12265	12955	12970	12983	12985	13001	13729	13747	13762	13764	13780	14556	14572	14588	14590	14608
	S/T	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.80	0.49	0.58	0.66	0.66	0.75	0.47	0.55	0.62	0.62	0.71
	PI	0.60	0.60	0.60	0.60	0.60	0.64	0.64	0.64	0.64	0.64	0.67	0.68	0.68	0.68	0.68	0.72	0.72	0.72	0.72	0.72
82°F (27.8°C)	TC	12379	12396	12409	12410	12426	13125	13141	13153	13156	13172	13909	13927	13943	13945	13961	14747	14763	14779	14782	14800
	S/T	0.56	0.66	0.75	0.75	0.85	0.53	0.62	0.70	0.70	0.80	0.50	0.58	0.66	0.66	0.76	0.47	0.55	0.63	0.63	0.71
	PI	0.67	0.67	0.67	0.67	0.67	0.71	0.71	0.71	0.71	0.71	0.75	0.75	0.76	0.76	0.76	0.80	0.80	0.80	0.80	0.80
86°F (30.0°C)	TC	12695	12712	12726	12726	12744	13460	13476	13489	13492	13508	14264	14283	14299	14301	14317	15124	15140	15157	15159	15178
	S/T	0.58	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79	0.49	0.58	0.65	0.65	0.75
	PI	0.81	0.81	0.81	0.81	0.81	0.86	0.86	0.86	0.86	0.86	0.91	0.91	0.91	0.91	0.91	0.96	0.96	0.97	0.97	0.97
95°F (35.0°C)	TC	13406	13424	13439	13440	13458	14214	14231	14245	14248	14265	15064	15083	15100	15103	15120	15971	15989	16006	16008	16028
	S/T	0.64	0.75	0.85	0.85	0.97	0.60	0.71	0.81	0.81	0.92	0.57	0.67	0.76	0.76	0.87	0.54	0.63	0.72	0.72	0.82
	PI	1.12	1.12	1.12	1.12	1.12	1.19	1.19	1.19	1.19	1.19	1.2									

Climate 5000 Series - Gen 4

12K Btu Single Zone Max Performance System - 4-Way Cassette IDU



BOSCH

Heating Performance - Indoor Dry Bulb Temperature at 418 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	4410	4322	4300	4283	4244	1.20	1.21	1.21	1.21	1.22
-13°F(-25.0°C)	5743	5628	5600	5578	5527	1.25	1.26	1.26	1.27	1.27
-4°F(-20.0°C)	7692	7538	7500	7470	7403	1.31	1.32	1.32	1.33	1.33
0°F(-17.8°C)	9515	9324	9278	9241	9158	1.47	1.48	1.48	1.49	1.49
5°F(-15.0°C)	11794	11558	11500	11454	11351	1.67	1.68	1.69	1.69	1.70
14°F(-10.0°C)	12870	12613	12550	12500	12387	1.73	1.74	1.75	1.75	1.76
17°F(-8.3°C)	13229	12965	12900	12848	12733	1.75	1.76	1.77	1.77	1.78
22°F(-5.6°C)	13898	13620	13552	13498	13377	1.73	1.74	1.74	1.75	1.75
27°F(-2.8°C)	14567	14276	14205	14148	14020	1.70	1.72	1.72	1.72	1.73
32°F(0.0°C)	15236	14932	14857	14797	14664	1.68	1.69	1.69	1.70	1.70
35°F(1.7°C)	15638	15325	15248	15187	15051	1.66	1.68	1.68	1.68	1.69
37°F(2.8°C)	15817	15501	15424	15362	15224	1.61	1.62	1.62	1.62	1.63
42°F(5.6°C)	16267	15941	15862	15798	15656	1.46	1.47	1.48	1.48	1.48
44.6°F(7.0°C)	16447	16118	16037	15973	15829	1.40	1.41	1.42	1.42	1.43
47°F(8.3°C)	16716	16382	16300	16235	16089	1.32	1.33	1.33	1.33	1.34
52°F(11.1°C)	16455	16125	16045	15981	15837	1.17	1.18	1.18	1.18	1.19
57°F(13.9°C)	16193	15869	15790	15726	15585	1.02	1.03	1.03	1.03	1.03
59°F(15.0°C)	16088	15766	15688	15625	15484	0.96	0.97	0.97	0.97	0.97
62°F(16.7°C)	15931	15613	15534	15472	15333	0.87	0.88	0.88	0.88	0.88
64.4°F(18.0°C)	15806	15489	15412	15350	15212	0.80	0.80	0.81	0.81	0.81
69.8°F(21.0°C)	15523	15212	15136	15076	14940	0.64	0.64	0.64	0.64	0.65
75.2°F(24.0°C)	15240	14936	14861	14801	14668	0.47	0.48	0.48	0.48	0.48

Bolded values were tested. All other values are simulated and for reference only.